

**LATERAL STABILITY OF BEAMS AND
FRAMES****Section: 3.1 – STRENGTH OF METALLIC PARTS****Chapter: 3170****Title: LATERAL STABILITY OF BEAMS AND
FRAMES****Revision: 0.3****Date: 03/10/2022****1 INTRODUCTION**

Bending beams under certain conditions of loading and restraint can fail by lateral buckling in a manner similar to columns loaded in axial compression.

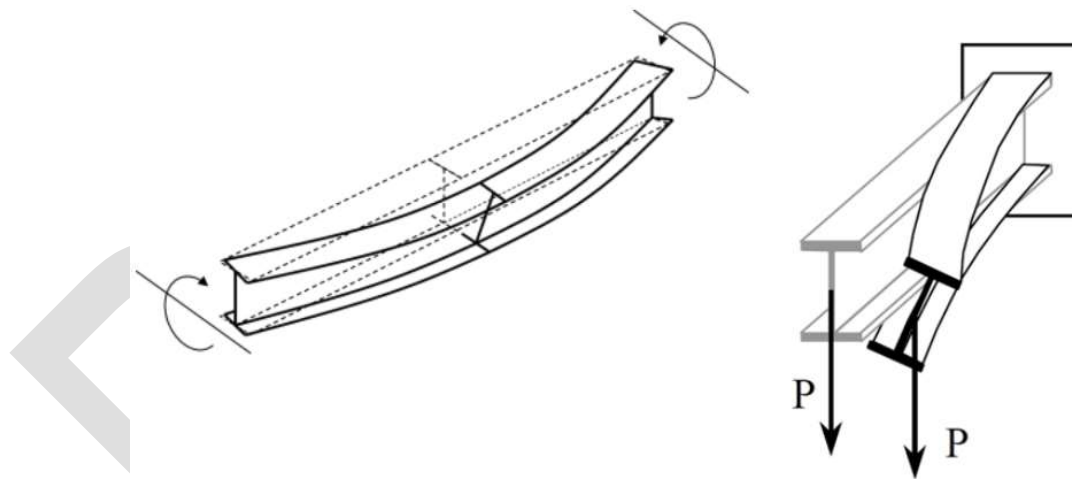


Figure 1. Lateral buckling of beams in bending

A typical conservative approach of this problem is to consider the compression side of the beam as a column. It is conservative as it neglects the torsional rigidity of the beam.

Solutions for some types of sections, loads and support conditions can be found in referenced bibliography, as in books from Niu (§10.5) or Roarks (Table 15.1, cases 11-16), or in HSB 44111-01. Details will not be shown herein.

2 LATERAL STABILITY OF FRAMES

A particular case of this type of failure may occur in stiffening members of a panel, in particular in case these stiffeners consist of profiles of open-thin-walled sections of significant height, which is the typical configuration of fuselage frames.

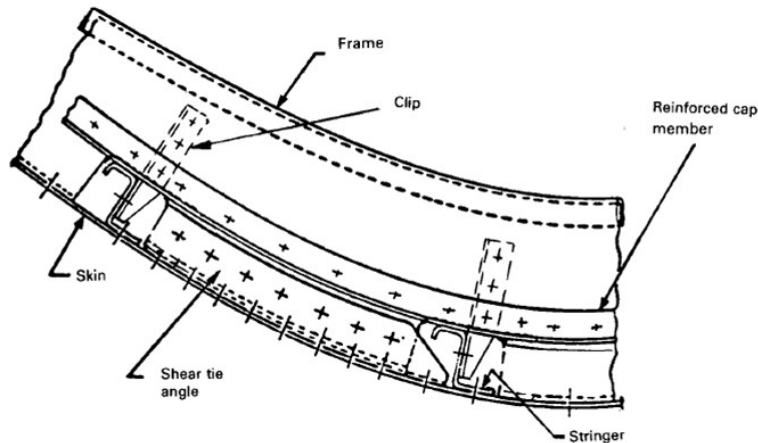


Figure 2. Typical Frame of a fuselage

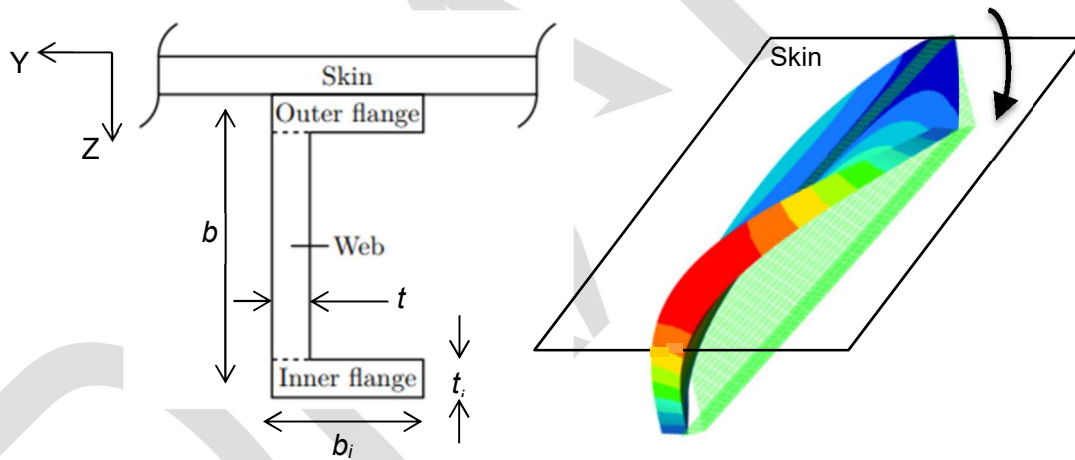


Figure 3. Frame+Skin section and lateral buckling

No closed-form solutions are known for this problem. Some approximations are proposed in following pages for approach sizing and stress justification of fuselage frames.

2.1 LATERAL BUCKLING OF FRAME SPANS BETWEEN STRINGERS

The most comparable solution found, up today, in technical literature, for this problem, is that proposed by M.C. Niu, in chapter 14.1 "SKIN-STRINGER PANELS", section "(E) LATERAL INSTABILITY OF OUTSTANDING STRINGER FLANGES" of his book (Ref. 1). For that case, the stiffened skin section is considered uniformly loaded in compression instead of bending, the typical critical case of ring frames. However, for critical bending cases of frames, as the neutral axis of the

LATERAL STABILITY OF BEAMS AND FRAMES

DEFENCE AND SPACE

Part:3.1
Chapter: 3170
Revision: 0.3
Date: 03/10/2022
Page 3 of 9

section shall be placed close to the skin, the piece of material relevant for the lateral instability problem, i.e. the most inner flange and its vicinities shall be as well loaded in compression, and Niu's solution is considered applicable.

Niu's proposal is to treat lateral instability problem as a beam (of section that shaded in following figure) on an elastic support.

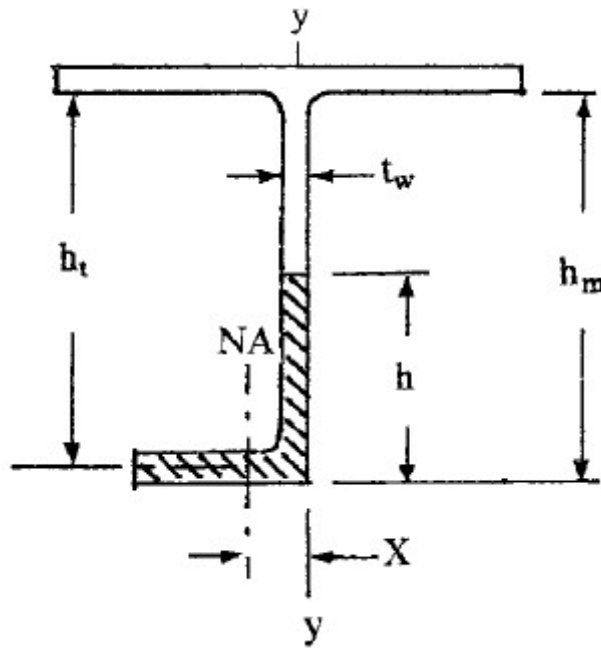


Figure 4. Beam section in lateral instability analysis: flange and effective web (Niu's Fig. 14.4.10)

The end supports of this beam, for the frame case, are the stringers positions, where it is expected that some stiffening members (clips in Figure 2) prevent the section to laterally buckle. The stringers pitch is then the length L of this straight beam (fuselage curvature is ignored for this short arc length). The analysis process proposed by Niu to get the allowable compression stress for lateral instability failure, now adapted to the frame problem, is as follows:

- 1) Calculate $\beta = E \cdot t_w^3 / (4 \cdot h_t^3)$
- 2) Obtain effective web height h from Fig. 14.4.11
- 3) Calculate section properties of the shaded area (area, position of neutral-axis / CG, bending moment of inertia and radius of giration) $A, X_{na}, I_{yy}, \rho_{yy}$
- 4) Calculate $\beta \cdot L^4 / (16 \cdot E \cdot I_{yy})$ and determine L'/L ratio from Fig. 14.4.12
- 5) Calculate the effective column length as $L' = (L'/L) \cdot L$

At this point Niu suggests to calculate the column buckling stress, including the appropriate plastic correction, by means of a graphical procedure...

- 6) Calculate $\epsilon = \pi / (L' \rho_{yy})^2$, note that $\rho_{yy} = (I_{yy} / A)^{0.5}$

LATERAL STABILITY OF BEAMS AND FRAMES

- 7) Read the allowable stress from an appropriate stress-tangent modulus curve (Ref. 1, figures 14.4.6 to 14.4.8), entering with parameter ϵ

Instead of that, for facilitating the implementation of the method and to overcome the limitation of materials for which a curve is available, the column buckling methodology discussed in chapter §3140 "COLUMN BUCKLING IN COMPRESSION" is recommended, applied on the simply supported beam of length L' and section that of the shaded area in previous discussion.

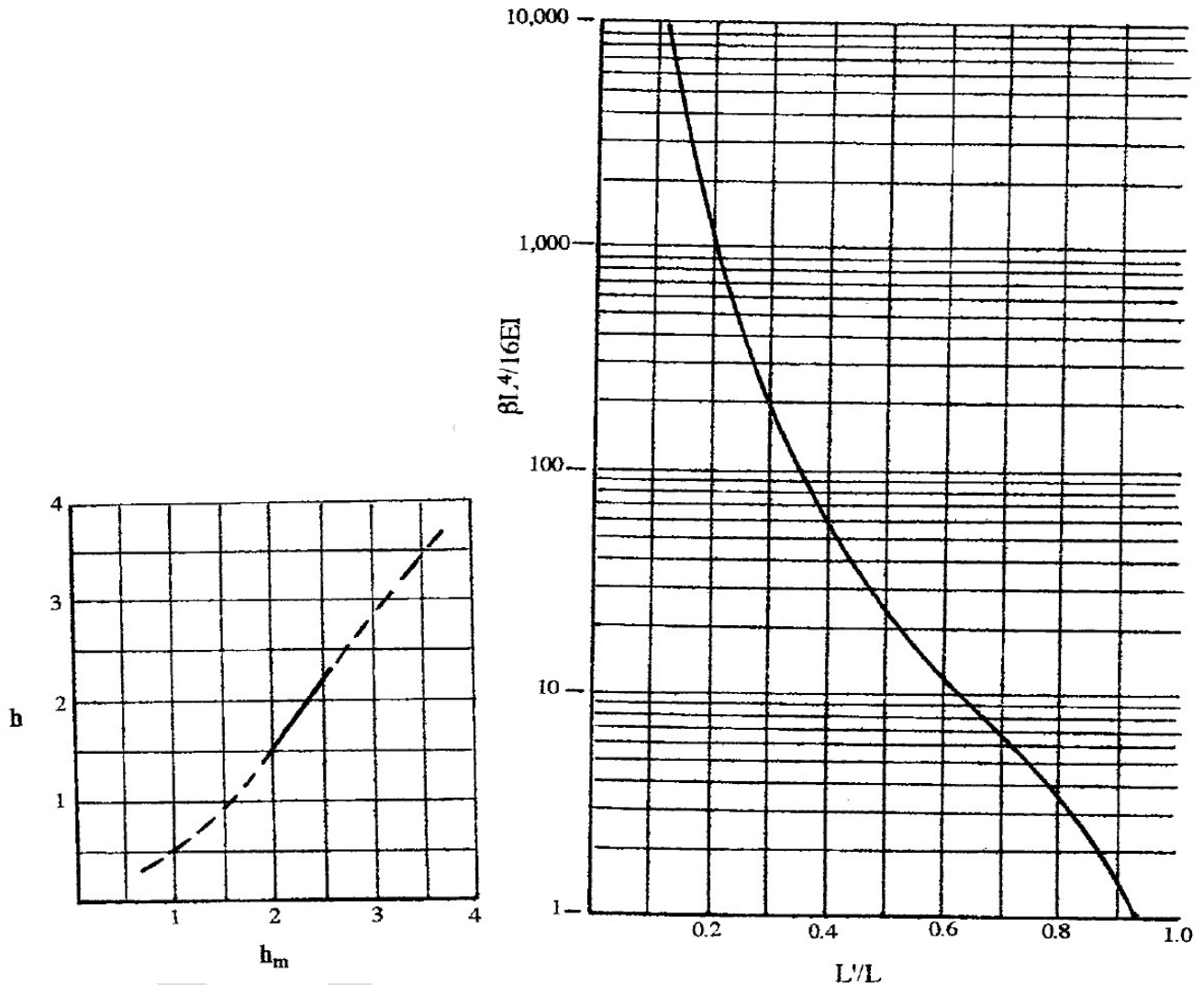


Figure 5. Effective web height and effective column length for lateral instability analysis

An alternative approximation to this problem is to use formulation proposed in HSB 44111-01. Assuming section warping is not prevented at end supports, the expression for the critical compression stress at the most inner fibre is:

$$\sigma_{cr-i} = 3.14 \cdot (E \cdot I_z \cdot G \cdot J)^{0.5} / (L \cdot W_y) = 3.14 \cdot (E \cdot I_z \cdot G \cdot J)^{0.5} / (L \cdot 2 \cdot I_y / H)$$

For previous analysis, the inertia properties shall be computed for the frame section taking into account that I_y has to be calculated relative to an axis at the skin line.

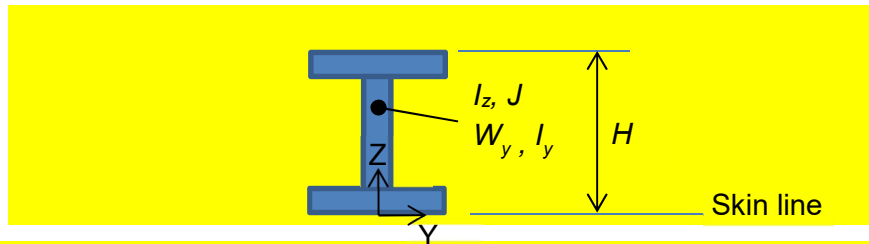


Figure 6. Effective frame section for computing inertias with HSB formulas for lateral buckling

(*) Note that conventions for designating reference axis systems in this last alternative and Niu's method are different.

2.1.1 Effectiveness of end stabilizers/clips

Method to be developed.

2.2 LATERAL BUCKLING OF UNSUPPORTED FRAMES

In this case support condition at skin and the web stiffness in out-plane bending and in torsion are the principal means of preventing the lateral buckling, as the L term in the previous buckling equations makes the contribution of the inner flange bending stiffness negligible.

The critical stress at the inner flange shall be obtained as a function approximation of FEM solutions for a battery of configurations representative of the design space, normalized as follows:

$$\sigma_{cr-i} / E = \text{function}(t/b, j_i)$$

...where t and b are the thickness and height of the web, E the Young modulus, and j_i a non-dimensional parameter representing the torsion stiffness of the inner flange/lip, e.g. $j_i = (b_i \cdot t_i^3) / (b \cdot t^3)$

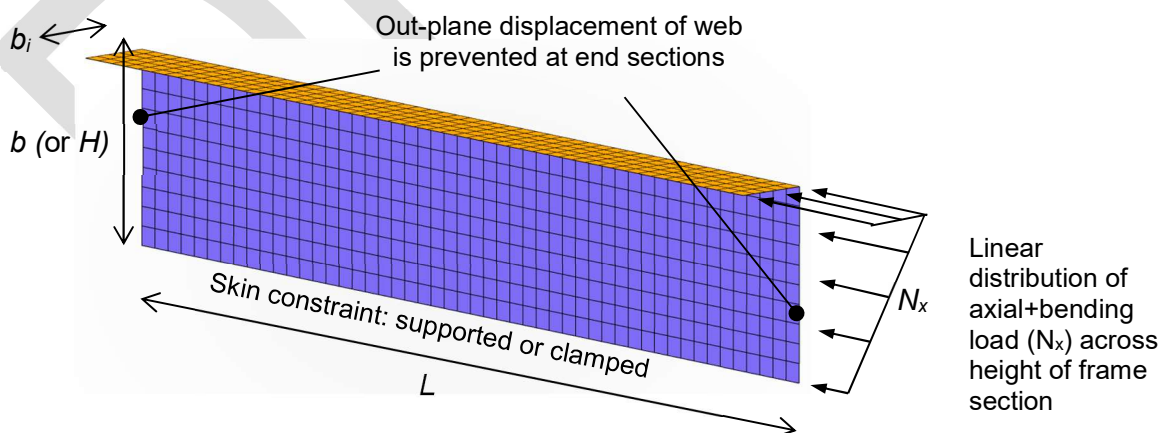


Figure 7. Description of FEM intended to be used for approximation of lateral buckling

The frame web rotation at skin support shall be considered prevented in this analysis. The justification for this assumption is that a section of a curved shell with a significant length in arc direction (curved skin) has a much greater bending stiffness than the plane frame web.

The approximation function shall be obtained for a reference load case consisting of a linear stress distribution with zero stress at skin support and reaching its maximum compression stress at the inner flange.

2.2.1 Lateral buckling of sheet folded long frame

Frames made of a metal folded sheet, characterized by an inner cap with just one flange, of the same thickness as the web t , and a width b_i , can be calculated with a FEM as that described above, using a length between end supports sufficiently big. Results for that configuration (rotation prevented at skin attachment) are summarized in following graph.

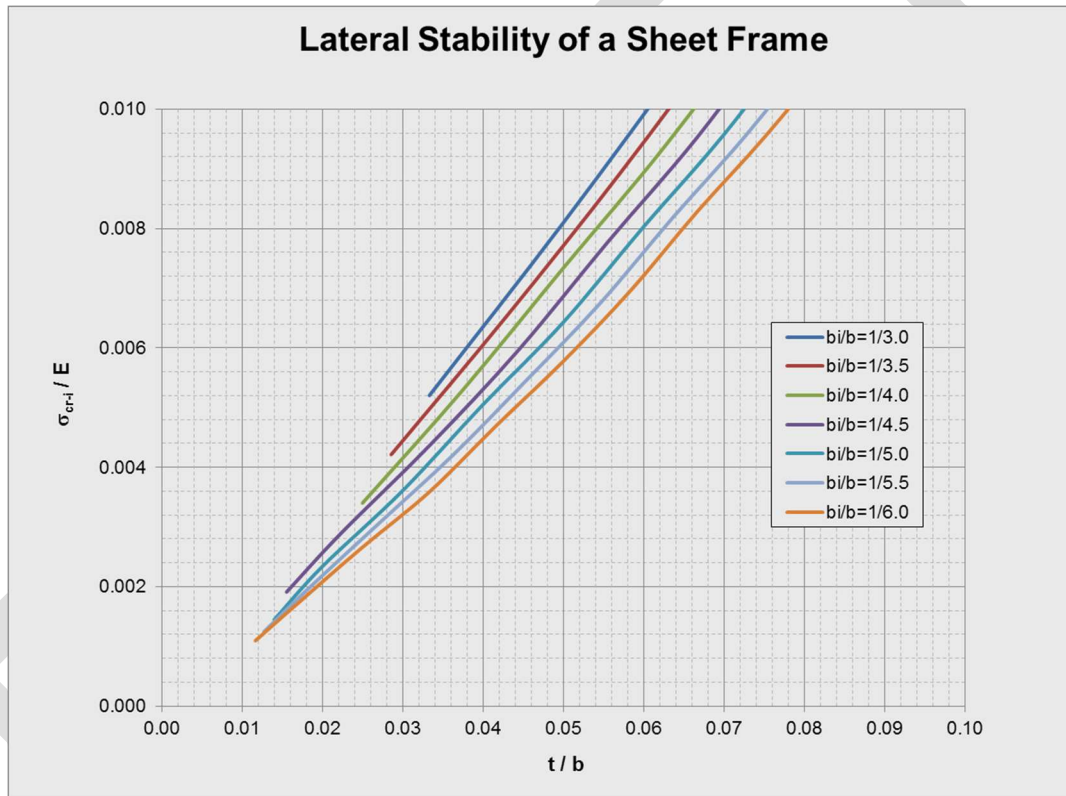


Figure 8. Critical lateral buckling inner flange compression stress of a folded sheet frame ($t_i=t$)

Those results can be approximated, within a range of width to height ratio b/b between 1/6 and 1/3, and thickness to height ratio t/b between 0.02 and 0.2 (*) as:

$$\sigma_{cr-i} / E = K_2 \cdot (t/b)^2 + K_1 \cdot (t/b)$$

...with:

$$K_2 = 1.0391 \cdot e^{-4.528 \cdot (b_i/b)}$$

$$K_1 = 0.0827 \cdot \ln(b_i/b) + 0.244$$

(*) The observed error within this application range is below 5%.

**LATERAL STABILITY OF BEAMS AND
FRAMES****2.2.2** **Lateral buckling of machined long frame**

Method to be developed.

DRAFT

DEFENCE AND SPACE

LATERAL STABILITY OF BEAMS AND FRAMES

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
A	mm ²	Cross-section area
b	mm	Width of section segment (web or flange)
cr	-	Abbreviation of critical, as subscript
E	MPa	Young modulus
H	mm	Height of frame web (width symbol b is also used for this purpose)
i	-	As subscript, refers to the inner cap
J	mm ⁴	Torsion moment of inertia of the cross-section
I	mm ⁴	Bending moment of inertia of the cross-section
L	mm	Length of beam
t	mm	Thickness of section segment (web or flange)
TBC	-	To Be Confirmed
W	mm ³	Bending modulus of the cross section, i.e. bending moment of inertia divided by the distance from neutral axis to the most external fibre of the section
Y,y	-	Direction of a local reference in cross section parallel to the skin surface
Z,z	-	Direction of a local reference in cross section parallel to the beam web
σ	MPa	Stress

REFERENCES

	Document Reference	Issue	Title
Ref. 1	[M. C. Y. Niu]	2nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 2	[W. C. Young & R. G. Budynas]	7th Ed.	ROARK'S FORMULAS FOR STRESS AND STRAIN
Ref. 3	HSB 44111-01	Is. B	Lateral buckling of deep beams

SOFTWARE

Program	Version	Description
NASTRAN (MSC-Software)	2012	Multidisciplinary structural analysis software based in the Finite Elements method
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures

Table 1. Software programs referenced in this chapter

LATERAL STABILITY OF BEAMS AND FRAMES

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.3 03/10/2022	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEPAP-TL4)</i>	G. Baños 03/10/2022	Florian Glock XX/XX/2022

Table 3. Approval signatures table for last revision of this chapter